

UDAY PRAKASH RASTOGI

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Summary

Pre-Final Year B.Tech Computer Science student passionate about Artificial Intelligence, Generative AI, and Software Engineering. Skilled in developing intelligent applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, AI Agents, and modern AI frameworks. Strong foundation in Data Structures & Algorithms, problem-solving, and scalable software development, with hands-on experience building AI-powered systems, automation workflows, and MERN-stack applications. Driven to create innovative, real-world solutions by combining cutting-edge AI technologies with strong engineering principles.

Education

Oriental Institute Of Science And Technology

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 7.6/10.0

Experience

TechnoHacks EduTech

Frontend Development Intern

Sep 2025 – Dec 2025

Remote

- Developed responsive web applications and interactive UI components using HTML, CSS, JavaScript, and React.js.
- Built projects including portfolio websites, landing pages, quiz applications, weather app interfaces, responsive navigation bars, and e-commerce layouts.
- Implemented responsive design principles, CSS animations, Flexbox/Grid layouts, and component-based architecture to enhance user experience across multiple devices.
- Optimized frontend performance and improved UI/UX using modern web development practices and reusable components.

CodSoft

C++ Programming Intern

Aug 2025 – Sep 2025

Remote

- Developed console-based applications including Library Management System, Number Guessing Game, Simple Calculator, Tic-Tac-Toe Game, and To-Do List Manager using C++.
- Applied Object-Oriented Programming concepts, file handling, modular programming, and problem-solving techniques to build scalable solutions.
- Strengthened understanding of Data Structures & Algorithms, debugging, and optimized coding practices through hands-on development tasks.

Google

Campus Ambassador

Aug 2025 – Present

Remote

- Promoted Google developer programs and organized student engagement activities at the campus level.
- Created technical outreach content and facilitated communication between students and developer communities.
- Collaborated with peers to increase awareness of Google technologies, developer tools, and learning initiatives.

Projects

AI Career Copilot (SkillSync AI) | *React.js, Node.js, Express.js, MongoDB Atlas, Google Gemini API, JWT, Google OAuth 2.0*

- Developed an AI-powered Career Copilot platform enabling ATS resume analysis, job matching, AI mock interviews, cover letter generation, and personalized career guidance through a unified SaaS application.
- Integrated Google Gemini API to deliver ATS scoring, skill-gap analysis, career coaching, interview question generation, and AI-powered resume enhancement recommendations.
- Built secure authentication and user management using JWT, Google OAuth 2.0, email verification, password reset workflows, and role-based protected routes.
- Designed scalable REST APIs and optimized database operations for managing resumes, interview reports, user profiles, and career insights efficiently.

PDF RAG Chatbot | *Python, LangChain, ChromaDB, Google Gemini, RAG, Streamlit*

- Built a Retrieval-Augmented Generation (RAG) chatbot capable of answering user queries from uploaded PDF documents with context-aware responses.
- Implemented document parsing, chunking, embedding generation, semantic search, and vector retrieval pipelines using LangChain and ChromaDB.
- Integrated Google Gemini LLM to provide accurate question answering, contextual reasoning, and conversational interactions over custom knowledge bases.
- Improved response quality through similarity-based retrieval and optimized embedding workflows for efficient information extraction.

LEGALBOT – AI Law Advisor Chatbot | *Python, NLP, LangChain, Google Gemini, RAG, Streamlit*

- Developed an AI-powered legal assistant capable of providing legal guidance, procedural information, and law-related query resolution through natural language conversations.
- Leveraged Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to deliver accurate and context-aware legal responses.
- Designed intelligent retrieval pipelines for legal knowledge access, improving relevance and reliability of generated answers.
- Built an interactive chatbot interface enabling users to understand legal rights, documentation requirements, and legal processes efficiently.

AI Learning Platform | *React.js, TypeScript, Firebase, Gemini API, Tailwind CSS*

- Developed a comprehensive AI-powered learning platform featuring learner, instructor, and admin dashboards with role-based access control.
- Implemented personalized learning recommendations, AI-assisted educational workflows, progress tracking, and intelligent assessment features.
- Integrated Firebase Authentication, Firestore, and cloud services for secure user management, course delivery, and achievement tracking.
- Designed a scalable and responsive architecture supporting interactive video courses, learning analytics, and real-time user engagement.

Technical Skills

Programming Languages: Python, C++, JavaScript, TypeScript, SQL

AI & Generative AI: Generative AI, Agentic AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Prompt Engineering, AI Agents, Vector Databases, Embeddings

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Tailwind CSS

Databases & Cloud: MongoDB, Firebase, ChromaDB, SQL

Developer Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, REST APIs, Problem Solving

Achievements & Awards

LeetCode: Solved 300+ coding problems covering Data Structures, Algorithms, and Problem Solving – [View Profile](#).

GeeksforGeeks: Solved 110+ coding problems focused on Data Structures, Algorithms, and Competitive Programming – [View Profile](#).

Research Publication

An Advanced AI-Powered Legal Advisor Chatbot for Rural India : Published in International Journal of Research Publication and Reviews (IJRPR), Vol. 6, Issue 12, Dec 2025. [View Publication](#)

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate [View Certificate](#)

Generative AI for All – Infosys Springboard [View Certificate](#)

Fundamentals of Artificial Intelligence – Code360 [View Certificate](#)

SQL for Data Analysis – LinkedIn Learning [View Certificate](#)

Skill Up with Python: Hands-On Data Science and Machine Learning Projects – LinkedIn Learning [View Certificate](#)

Frontend Development Certification – OneRoadmap [View Certificate](#)

React Bootcamp – LetsUpgrade [View Certificate](#)

Ethical Hacker – Cisco Networking Academy [View Certificate](#)

Data Analytics Job Simulation – Deloitte [View Certificate](#)